Generalities

Tout replier

Advertising

Groups

Repartition of students in group 1 and Group 2
Group1 : Français/French
Group2: Anglais/English

Python For Instrumentation and Laboratory experiments - Pr. Thierry Angot

Computer Room (2nd floor) in **A. Perot Building** in St-Jérôme campus

2 groups: Gr1 8h30-12h00 and Gr2 13h30-17h00; Tuesday 08/09 , 15/09, 29/09

Computer Management - Pr. Bruno Denet

2 groups: Tuesday 08/09, 15/09 : Gr2 8h30-12h00 Gr1 13h30-17h00 salle bleue,

06/10 : Gr1 8h30-12h00 online (previously salle bleue) Gr2 13h30-17h00 online (previously salle verte);

Sesao Computer Room in St-Jérôme campus. Entry BJ2, close to the cafeteria, last floor.

zoom meeting computer management

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Map of the saint-Jérôme campus

Python for Instrumentation and Laboratory Experiments

This class aims at providing the bases for programming with Python under Jupiter Notebook environment, and further to understand and to interface by Serial or Parallel communication with Instruments.

A short course on Python

Basics of communication

Keithley 2000 manual

Keithely 2000 - Digest

Acquisition of temperature with a thermocouple

Programming a leak valve

Dear students,

This task consist in writing a Python program in order to control a leak valve through its RS232 port. You will need to import the appropriate librairies, test the device status, check the valve position, set to a chosen (but arbitrary) position, hold it in position for a certain amount of time, and finally close it. Finally you will write a report about the functioning of the valve and documenting your Python program.

To this end you first need to read the documentation of the valve, understand how it works and how to communicate with it.

You will zip these files and upload them into the 'drop box' before Sunday 10th of October.

Regards,

Thierry Angot

Drop box

Ouvert le : mardi 29 septembre 2026, 08:00

Before 11h of October midnight, you need to provide a zip file with 4 Jupyter Notebook.
1. Summary of the first class, and solution of the exercices proposed in the short course on Python
2. Notebook of your exercises communicating with the Keithley using pyserial.
3. Notebook with temperature acquisition with the Keithley 2000, you will need for Experimental Physics: Interfacing a thermocouple to record the temperature evolution : surfusion of tin
4. Notebook on how you would imagine to communicate with the leak valve.

Computer management

In this class, you will learn some bases of unix command line, make, git version control system, latex.

You can either use the virtualized desktops of the sesao computer rooms, or preferably your own computer.

If you use windows, install wsl, which involves essentially typing wsl --install as administrator, possibly rebooting. Then you have an ubuntu distribution inside windows, you install packages by

sudo apt install some_package

make can be found in the package build-essential

You can either install latex on your computer (package texlive-full on ubuntu or wsl) which can be very large or use overleaf (overleaf.com), an online latex editor. In the end, it will be mandatory to create a final report with latex and bibtex. Use figures (some screenshots), labels and cross-references. You will then send it by mail (bruno.denet@gmail.com) before November 15.

- How to install wsl
- Introduction to Unix command line
- Introduction to make and Makefile
- Short introduction to latex (part 1)
- Short introduction to latex (part 2)
- Introduction to git
- git main commands
- A book on git (look at the first chapters)

Nouvelle section

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